BigQuery Omni — No-Copy Architecture

How GCP BigQuery queries Oracle data in your cloud — without moving a single byte to Google Cloud

AWS or Azure

① Oracle Database

Source of truth: members · claims · prior_auth
Runs on EC2 (AWS) or Azure VM

② oracle_to_iceberg.py (PyIceberg)

Extracts rows → converts to Parquet → creates Iceberg snapshots with metadata

③ Object Storage (S3 or Azure Blob)

healthcare/members/data.parquet
healthcare/claims/data.parquet

④ Iceberg REST Catalog

Schema + snapshot metadata
Points to Parquet locations · Backed by PostgreSQL

⑤ BigQuery Omni Compute (runs INSIDE your cloud VPC) | Reads storage directly | Never sends raw data to GCP

NO raw data →

Results →

← Query

Google Cloud

⑥ Secret Manager

Stores IAM Role ARN (AWS) or Service Principal (Azure)
+ Iceberg catalog URL

⑦ BigQuery Connection

Lakehouse on source cloud (via BigQuery Omni)
Assumes IAM Role via OIDC federation

⑧ BigQuery External Tables

healthcare_lakehouse.members
healthcare_lakehouse.claims

⑨ BigQuery SQL Query

SELECT m.first_name, SUM(c.billed_amount)
FROM members m JOIN claims c ...

→ Results returned. Oracle data never left your cloud.

Oracle data never crosses cloud boundary • BigQuery Omni compute runs inside your VPC • Only query + aggregated results travel to GCP

How No-Copy Works — What Travels vs What Stays

Traditional ETL copy pipeline vs BigQuery Omni federated query • AWS or Azure

❌ Traditional — What crosses the cloud boundary

Oracle extracts ALL rows

Full table dump — every PHI record copied

ETL transforms data

ADF / Glue / Dataflow reshapes the data

ALL data copied to GCS

Complete PHI dataset lands in GCP storage

BigQuery reads from GCP

Queries run on GCP — data already duplicated

Repeat nightly

Pipeline runs again — another full copy

VS

✅ BigQuery Omni — What crosses the cloud boundary

SQL query text only →

SELECT statement sent to Omni node (tiny payload)

Omni reads storage locally

Parquet files read INSIDE your cloud — no cross-cloud transfer

Filter + aggregate in your cloud

WHERE, GROUP BY, SUM run inside your VPC

Only results ←

Aggregated result rows returned to GCP (kilobytes, not GB)

Data never moves

Oracle / Storage PHI stays 100% inside your cloud boundary forever

0 bytes
of PHI transferred
to Google Cloud

100%
of compute runs
inside your VPC

1
PHI boundary
for HIPAA audit

Real-time
always current
no ETL lag

AWS vs Azure — BigQuery Omni Setup Differences

Same architecture, different cloud primitives — the BigQuery SQL queries remain identical

Component	AWS	Azure
Oracle Host	EC2 t3.small (us-east-1)	Azure VM (any region)
Object Storage	AWS S3 Bucket	Azure Blob Storage
Iceberg Catalog	Iceberg REST + PostgreSQL on EC2	Iceberg REST + PostgreSQL on Azure VM
Auth to GCP	IAM Role ARN via OIDC federation	Service Principal via OIDC / OAuth
BQ Connection	BigQuery Omni connection (AWS)	BigQuery Omni connection (Azure)
Secret Storage	GCP Secret Manager stores IAM Role ARN	GCP Secret Manager stores SP credentials
External Table	FORMAT='ICEBERG', uris=[s3://...]	FORMAT='ICEBERG', uris=[abfs://...]
BQ Compute Region	AWS us-east-1 (Omni node)	Azure eastus (Omni node)

✅ **Your SQL queries are 100% identical on both clouds — only the connection string and auth method differ**